I. System Model and Problem Formulation

A. System Model

1) Network Scenario: We consider a OneWeb-like LEO satellite communication system and focus on downlink interference management under NGSO–GSO coexistence. Each LEO satellite is equipped with 16 spot beams [13], and serves terrestrial users within its coverage area in every time slot.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t, and let $\mathcal{B}_s$ denote the set of beams on satellite s. We assume that every LEO satellite has the same number of beams:

$$|\mathcal{B}_s| = 16, \qquad \forall s \in \mathcal{S}(t).$$

Each satellite employs a fixed beam layout with 16 spot beams arranged along the along-track (north–south) direction. For beam b on satellite s, the ground footprint at time slot t is denoted by

$$\mathcal{F}_{s,b}(t).$$

If user u lies in $\mathcal{F}_{s,b}(t)$, user u is covered by beam b on satellite s and may be considered as a candidate for subsequent user association.

The system also includes a protected GSO ground station, denoted by g. When a visible LEO satellite passes close to the GS boresight or forms a near-inline geometry with respect to the GSO direction, its downlink transmission may produce a high aggregate EPFD at the protected GS. We define the LEO satellite that poses the dominant EPFD risk to the protected GS at time slot t as the *critical satellite*:

$$s^{\mathrm{c}}(t).$$

We further consider the along-track north and south neighbors of the critical satellite as candidate helper satellites for subsequent service transfer and relaying:

$$s^{\mathrm{N}}(t), \quad s^{\mathrm{S}}(t).$$

where $s^{\mathrm{N}}(t)$ and $s^{\mathrm{S}}(t)$ denote the north and south helper satellites, respectively. Because the considered LEO constellation follows a regular Walker-star orbital structure, the relative geometry and beam coverage overlap among the critical satellite and its helpers are predictable. This property underpins the beam-level user reassociation and relay design in the next chapter.

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of users within the system service region at time slot t. Each user can be served by at most one beam on one LEO satellite in the same time slot. The user–beam association is represented by

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$.

The set of users served by beam b on satellite s is

$$\mathcal{U}_{s,b}(t) = \{u \in \mathcal{U}(t) \mid x_{u,s,b}(t) = 1\}.$$

Hence,

$$N_{s,b}(t) = \sum_{u \in \mathcal{U}(t)} x_{u,s,b}(t). \tag{1}$$

Because each user is served by only one beam, the downlink rate of a user is determined solely by its serving beam. The service quality of a user depends on the link condition, beam power, channel assignment of the serving beam, and the number of co-served users on that beam.

If user u is not within the footprint $\mathcal{F}_{s,b}(t)$ of beam s, b, that beam is not a feasible serving beam for user u. Association decisions are therefore made only among satellite–beam pairs that can cover the user, ensuring that subsequent association, reassociation, and relay decisions respect actual beam coverage.

3) Beam Power Model: We adopt a baseline-power beam operation model. In normal operation, when LEO satellites are far from the protected GS and the EPFD violation risk is low, all active beams on each satellite are configured with the same baseline power. For beam b on satellite s, the transmit power is denoted by

$$P_{s,b}(t).$$

In normal operation,

$$P_{s,b}(t) = P^{\mathrm{base}}, \qquad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s.$$

where P^{base} is the baseline power under normal conditions. The purpose of baseline power is to provide a fixed configuration that meets typical user demand without solving a full per-time-slot continuous power control problem. Because the numbers of beams and users are large, re-solving a complete power control problem in every time slot is computationally demanding and difficult to deploy in practice. We therefore use baseline-power operation as the initial configuration in normal operation and concentrate subsequent power adjustments on EPFD-risk periods involving beam shutdown and helper beam power boosting.

Each beam has a maximum transmit power $P_{s,b}^{\max}$, and

$$0 \le P_{s,b}(t) \le P_{s,b}^{\max}, \qquad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s. \tag{2}$$

Each satellite also has a total transmit power limit $P_s^{\max}$:

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \le P_s^{\max}, \qquad \forall s \in \mathcal{S}(t). \tag{3}$$

When the critical satellite $s^{\mathrm{c}}(t)$ approaches the protected GS and creates an aggregate EPFD violation risk, some high-risk beams may be shut down. Let $\mathcal{B}_{\mathrm{c}}^{\mathrm{off}}(t)$ denote the set of beams switched off on the critical satellite. Then

$$P_{s^{\mathrm{c}},b}(t) = 0, \qquad \forall b \in \mathcal{B}_{\mathrm{c}}^{\mathrm{off}}(t).$$

The remaining beams are *safe beams*, denoted by $\mathcal{B}_{\mathrm{c}}^{\mathrm{safe}}(t)$, and keep the baseline power:

$$P_{s^{\mathrm{c}},b}(t) = P^{\mathrm{base}}, \qquad \forall b \in \mathcal{B}_{\mathrm{c}}^{\mathrm{safe}}(t),$$

where

$$\mathcal{B}_{\mathrm{c}}^{\mathrm{safe}}(t) = \mathcal{B}_{s^{\mathrm{c}}} \setminus \mathcal{B}_{\mathrm{c}}^{\mathrm{off}}(t).$$

Hence,

$$P_{s^{\mathrm{c}},b}(t) = \begin{cases} 0, & b \in \mathcal{B}_{\mathrm{c}}^{\mathrm{off}}(t), \\ P^{\mathrm{base}}, & b \in \mathcal{B}_{\mathrm{c}}^{\mathrm{safe}}(t). \end{cases}$$

For helper satellites, beams operate at baseline power under normal conditions. When a helper must take over users from closed beams on the critical satellite, relaying helper beams may be raised up to their beam-level maximum:

$$P_{s^{\mathrm{h}},b}(t) \le P^{\max}_{s^{\mathrm{h}},b}, \qquad s^{\mathrm{h}} \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\},$$

subject to (3). In the proposed method of the next chapter, critical safe beams may absorb part of the helper traffic so that the helper satellite can release power back to its available power pool and boost the relaying helper beams. Although transmit power may be reallocated among beams on the same helper satellite, each user is still served by only one beam, and the delivered service rate is determined only by the capacity of that serving beam.

4) SINR Model: The received quality of a user is determined by the desired signal from the serving beam, co-channel interference, and receiver noise. We adopt a OneWeb-like multi-channel frequency assignment with channel set $\mathcal{K}$ and

$$|\mathcal{K}| = 8.$$

If beam b on satellite s uses channel $k_{s,b} \in \mathcal{K}$, the user suffers co-channel interference only from beams that reuse the same channel as the serving beam.

If user u is served by beam b on satellite s, the received SINR is

$$\mathrm{SINR}_u(t) = \frac{S_u(t)}{I^{\mathrm{intra}}_u(t) + I^{\mathrm{inter}}_u(t) + N_u(t)}. \tag{4}$$

where $S_u(t)$ is the desired signal power, $I^{\mathrm{intra}}_u(t)$ and $I^{\mathrm{inter}}_u(t)$ are intra-satellite and inter-satellite co-channel interference, respectively, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: For user u served by beam b on satellite s, the desired signal power is

$$S_u(t) = x_{u,s,b}(t)\, \frac{P_{s,b}(t)\, G^{\mathrm{tx}}_{s,b\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s,u}(t)}.$$

where $P_{s,b}(t)$ is the transmit power of beam b, $G^{\mathrm{tx}}_{s,b\to u}(t)$ is the transmit gain of beam b toward user u, G^{rx}_u is the user receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u.

The total propagation loss is modeled as

$$L_{s,u}(t) = L^{\mathrm{fs}}_{s,u}(t)\, L^{\mathrm{gas}}_{s,u}(t),$$

where $L^{\mathrm{fs}}_{s,u}(t)$ and $L^{\mathrm{gas}}_{s,u}(t)$ denote free-space and atmospheric gas losses, respectively. The free-space path loss is

$$L^{\mathrm{fs}}_{s,u}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda}\right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u, and λ is the carrier wavelength.

Intra-Satellite Interference: Because different beams on the same satellite may reuse the same channel, a user may suffer interference from other co-channel beams on its serving satellite. This component is defined as intra-satellite interference.

If user u is served by beam b on satellite s, the intra-satellite interference is

$$I^{\mathrm{intra}}_u(t) = \sum_{\substack{b' \in \mathcal{B}_s \\ b' \neq b}} \mathbf{1}\big(k_{s,b'} = k_{s,b}\big)\, \frac{P_{s,b'}(t)\, G^{\mathrm{tx}}_{s,b'\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s,u}(t)},$$

where $\mathbf{1}(k_{s,b'} = k_{s,b})$ is a channel indicator function that equals 1 if beam b' reuses the serving channel and 0 otherwise.

Inter-Satellite Interference: In addition to intra-satellite interference, a user may be interfered with by co-channel beams on other visible satellites. This component is defined as inter-satellite interference.

If user u is served by beam b on satellite s, the inter-satellite interference is

$$I^{\mathrm{inter}}_u(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}} \mathbf{1}\big(k_{s',b'} = k_{s,b}\big)\, \frac{P_{s',b'}(t)\, G^{\mathrm{tx}}_{s',b'\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s',u}(t)},$$

where $G^{\mathrm{tx}}_{s',b'\to u}(t)$ is the transmit gain of interfering beam s', b' toward user u, and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u.

Under the OneWeb-like 8-channel architecture, the received quality depends on the serving beam power, antenna gain, and path loss, as well as co-channel intra-satellite and inter-satellite interference. Beam shutdown, user reassociation, and helper relaying can therefore change user-side SINR and satisfaction.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the instantaneous capacity follows from the Shannon formula. Let B_u denote the effective bandwidth of user u. The instantaneous capacity of user u at time slot t is

$$C_u(t) = B_u \log_2\!\big(1 + \mathrm{SINR}_u(t)\big).$$

Because multiple users may be served by the same beam, intra-beam resource sharing is modeled by TDMA. If user u is served by beam b on satellite s, the number of served users on the beam is $N_{s,b}(t)$. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}. \tag{5}$$

where $N_{s,b}(t)$ is defined in (1). Equation (5) indicates that users on the same beam share the downlink resource of that beam through TDMA. Because each user is served by only one beam, $R_u(t)$ is determined solely by the serving beam.

Let $D_u(t)$ denote the demand rate of user u at time slot t. The satisfaction of user u is defined as

$$\mathrm{sat}_u(t) = \min\!\left(\frac{R_u(t)}{D_u(t)}, 1\right). \tag{6}$$

where $\mathrm{sat}_u(t) \in [0,1]$. If $\mathrm{sat}_u(t) = 1$, the demand of user u is fully met; if $0 < \mathrm{sat}_u(t) < 1$, user u receives only partial service; if $\mathrm{sat}_u(t) = 0$, user u receives no effective service in the time slot.

This satisfaction metric is used in the problem formulation and performance evaluation. The goal is to maximize the total satisfaction of all users subject to EPFD and power constraints.

6) Beam-Level EPFD Model: Under NGSO–GSO coexistence, the LEO satellite system must keep the aggregate EPFD at the protected GSO ground station below the regulatory limit [5], [6]. Because each LEO satellite employs 16 spot beams with different pointing directions, off-axis angles, and transmit powers toward the GS, we adopt beam-level EPFD calculation and sum the contributions in the linear domain.

For beam b on satellite s, the EPFD contribution toward the protected GS g is defined as

$$\mathrm{EPFD}_{s,b}(t) = \frac{P_{s,b}(t)\, G^{\mathrm{tx}}_{s,b\to g}(t)}{4\pi d^2_{s,g}(t)} \cdot \frac{G^{\mathrm{rx}}_g(\theta_{g\to s}(t))}{G^{\mathrm{rx}}_{g,\max}} \cdot \frac{1}{B_{\mathrm{ref}}},$$

where $P_{s,b}(t)$ is the beam transmit power, $G^{\mathrm{tx}}_{s,b\to g}(t)$ is the transmit gain toward the GS, $d_{s,g}(t)$ is the satellite–GS distance, $G^{\mathrm{rx}}_g(\theta_{g\to s}(t))$ is the GS receive gain, $G^{\mathrm{rx}}_{g,\max}$ is the peak GS receive gain, $\theta_{g\to s}(t)$ is the off-axis angle from the GS toward satellite s, and B_{ref} is the reference bandwidth [7], [8].

The transmit gain toward the GS is expressed as a function of the off-axis angle:

$$G^{\mathrm{tx}}_{s,b\to g}(t) = G^{\mathrm{tx}}_{s,b}\big(\phi_{s,b\to g}(t)\big),$$

where $\phi_{s,b\to g}(t)$ is the off-axis angle from the boresight of beam b to the protected GS. Because the 16 beams on the same satellite point in different directions, their EPFD contributions must be computed individually.

The total EPFD contribution of satellite s is the linear sum over all beams:

$$\mathrm{EPFD}^{\mathrm{sat}}_s(t) = \sum_{b\in\mathcal{B}_s} \mathrm{EPFD}_{s,b}(t).$$

The aggregate EPFD from all visible LEO satellites at time slot t is

$$\mathrm{EPFD}_{\mathrm{agg}}(t) = \sum_{s\in\mathcal{S}(t)} \mathrm{EPFD}^{\mathrm{sat}}_s(t).$$

Equivalently,

$$\mathrm{EPFD}_{\mathrm{agg}}(t) = \sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} \mathrm{EPFD}_{s,b}(t).$$

To ensure regulatory compliance, the system must satisfy

$$\mathrm{EPFD}_{\mathrm{agg}}(t) \le \mathrm{EPFD}_{\mathrm{lim}}, \tag{7}$$

where $\mathrm{EPFD}_{\mathrm{lim}}$ is the applicable EPFD regulatory limit. If $\mathrm{EPFD}_{\mathrm{agg}}(t) > \mathrm{EPFD}_{\mathrm{lim}}$ in a time slot, the current beam power configuration produces excessive GS interference and must be adjusted through beam shutdown or user reassociation.

B. Problem Formulation

Based on the system model above, we maximize the total user satisfaction subject to user association, beam power, satellite total power, and EPFD constraints. The joint optimization problem is formulated as follows.

Variables: The optimization variables are the user–beam association indicators $\mathbf{x}(t) = \{x_{u,s,b}(t)\}$ and the beam transmit powers $\mathbf{P}(t) = \{P_{s,b}(t)\}$.

Objective function:

$$\max_{\mathbf{x}(t),\mathbf{P}(t)} \sum_{u\in\mathcal{U}(t)} \mathrm{sat}_u(t). \tag{8}$$

Constraints:

$$\sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} x_{u,s,b}(t) \le 1, \quad \forall u \in \mathcal{U}(t), \tag{9}$$

$$0 \le P_{s,b}(t) \le P^{\max}_{s,b}, \quad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s. \tag{10}$$

$$\sum_{b\in\mathcal{B}_s} P_{s,b}(t) \le P^{\max}_s, \quad \forall s \in \mathcal{S}(t). \tag{11}$$

$$\mathrm{EPFD}_{\mathrm{agg}}(t) \le \mathrm{EPFD}_{\mathrm{lim}}. \tag{12}$$

Constraint (9) ensures that each user is served by at most one satellite–beam pair in each time slot.

Constraint (10) limits each beam to $[0, P^{\max}_{s,b}]$, allowing shutdown (0) and helper boosting (up to $P^{\max}_{s,b}$).

Constraint (11) enforces the satellite total power budget $P^{\max}_s$.

Constraint (12) enforces the aggregate EPFD regulatory limit at the protected GS.

The objective in (8) maximizes the aggregate satisfaction of all users under the EPFD and power constraints. Because $\mathrm{sat}_u(t)$ is defined in (6), the objective reflects the overall service quality in time slot t.

Because association affects TDMA sharing, SINR, service rate, and satisfaction, while beam power affects both user-side link quality and GS-side aggregate EPFD, the problem is highly coupled. Near the protected GS, beam shutdown reduces EPFD but degrades users on closed beams; reassociating them to helpers increases helper beam loading and may hurt existing helper users.

We therefore do not pursue a global optimum of this mixed-integer nonlinear program. Instead, the next chapter presents a staged heuristic framework that satisfies the EPFD constraint through safe-beam assisted power release and beam-pair local relay to improve satisfaction during critical periods.

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